

Master Project Summary: Unlocking \$460M in Hidden Revenue

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1. The Strategic Vision (The "Why")

The Problem: Traditional banking underwriting is plagued by a "Conservative Bias." By rejecting borrowers in Grade D and E segments based on rigid scores, institutions suffer from a high **False Negative Rate**, leaving billions in safe, high-yield interest income on the table.

The Solution: A "Second-Look" AI Engine designed to identify "**Hidden Gems**"—borrowers who are statistically low-risk but mispriced by legacy systems.

2. Technical Execution Roadmap (The "How")

Phase A: Data Engineering at Enterprise Scale

- **Dataset:** 2.26 Million records from the LendingClub portfolio.
- **Optimization:** Implemented **Memory Downcasting** (converting float64 to float32) to process the entire portfolio within a 30GB cloud RAM environment.
- **Integrity:** Scrubbed "Post-Origination" variables to eliminate **Data Leakage**, ensuring the model only uses information available at the moment of application.

Phase B: Advanced Risk Modeling

- **Weight of Evidence (WoE):** Transformed categorical features into predictive risk scales.
- **Class Balancing (SMOTE):** Artificially balanced the dataset to ensure the model learned the subtle nuances of defaults (Class 1) despite their rarity.
- **The "Champion" Model:** Deployed an **XGBoost Classifier** (Gradient Boosting) for high-precision probability scoring.
- **The "When" (Survival Analysis):** Used the **Cox Proportional Hazards Model** to predict "Time-to-Default," distinguishing between toxic early defaults and profitable late-stage risk.

3. The Business Outcome (The "Results")

After running the calibrated model on the full 2.2M portfolio, the results were enterprise-shaping:

KPI	Achievement
Model Accuracy	91%
Recall (Catching Defaults)	83%
Hidden Gems Identified	153,307 Borrowers
Unlocked Loan Volume	\$2,460,865,280 (\$2.46 Billion)
ANNUAL UNLOCKED REVENUE	\$460,071,296 (\$460 Million)

4. Professional Assets Generated

To showcase this project to top-level recruiters, we have developed:

1. **Technical White Paper:** A professional PDF report detailing the methodology and strategic recommendations.
2. **GitHub Repository:** A structured "Sales Page" with clear commit history and technical documentation.
3. **LinkedIn Strategy:** A "Thought Leadership" approach using PDF carousels and "The \$460M Hook."
4. **Power BI Dashboard:** An interactive business intelligence tool to visualize the "Risk-Return Frontier."
5. **Portfolio Blueprint:** A website layout instructions guide that frames you as a growth-focused Product Manager.

5. Next Steps for Growth

- **Cloud Deployment:** Exploring the migration of this "Second-Look" engine into a real-time API.
- **Dynamic Pricing:** Recommending a "Risk-Based Repricing" strategy to reduce churn among the identified "Hidden Gems."